

Highlights

An Optimized Coverage Model for Wireless Visual Sensor Networks

- An overlap- and occlusion-aware coverage-degree (O^2ACD) model for wireless visual sensor networks.
- Per-target effective coverage as a union of visible arc intervals minus occluded sectors.
- Sensor-orientation planning solved with a tabular Double Q-Learning agent.
- Under matched evaluation budgets, Q-Learning outperforms PSO, DE, ACO, CA, and Harmony Search.

An Optimized Coverage Model for Wireless Visual Sensor Networks

ARTICLE INFO

Keywords:

Q-coverage model
coverage interval
target detection
wireless visual sensor networks
reinforcement learning
Q-Learning
Double Q-Learning

ABSTRACT

In this paper, an optimized coverage model is proposed for wireless visual sensor networks (WVSN), called the overlap and occlusion aware coverage degree (O^2ACD) model. It considers both possible interval overlap between the viewing sensors and the interleavedly occluded targets. Also, a precise observation model is used, taking into account all possible overlaps between the sensor's Field of View (FoV) and the target. For each target, O^2ACD calculates the effective coverage degree with regard to covering sensors, eliminating the intersection between their coverage degrees. The resulting model is more accurate than the previous coverage degree model. The new model is evaluated using a tabular reinforcement-learning planner based on Q-Learning (QL) alongside well-known metaheuristic approaches such as PSO, DE, ACO, CA, and HS. The simulation results indicate that O^2ACD yields a materially different coverage estimate than the previous CD/TCD models by removing double-counted overlaps and occluded arcs. In addition, under matched evaluation budgets QL attains the highest coverage in all simulation scenarios, with Harmony Search (HS) the strongest of the metaheuristic baselines.

1. Introduction

Wireless visual sensor networks (WVSNs) deploy directional cameras to monitor targets in a surveillance field. Because each sensor has a limited field of view (FoV), it typically observes only a portion of a target; sensors can, however, adjust their orientations to improve visibility. The central objective is an optimized coverage model that maximizes target visibility and overall coverage ratio while minimizing the number of required sensors—supporting applications such as target tracking, surveillance, target monitoring, and intruder detection.

Accurate visual coverage must account for both FoV overlaps and line-of-sight occlusions. The O^2ACD model rectifies two core weaknesses of classical CD/TCD measures: it computes, for each target, the union of visible arc intervals (eliminating double counting) and explicitly subtracts occluded sectors, yielding a physically consistent scalar objective for optimization. In the original study, sensor orientations were planned with population-based metaheuristics (e.g., Harmony Search), which offer strong baselines but treat the evaluator as a black box at every iteration.

We advance this line of work by coupling O^2ACD with reinforcement learning. Reinforcement learning has recently shown strong results for sensor-coverage and deployment control, spanning UAV-swarm area coverage [6, 16], adaptive wireless-sensor deployment [19, 25], directional-sensor and camera management [2, 4], and multi-agent vision-based coverage [21]. Beyond population-based search, we adopt tabular Q-Learning (QL) to learn a value function over orientation actions, guided by O^2ACD 's physically grounded signals—(i) marginal increases in union-of-arc coverage, (ii) penalties for redundant overlap, and (iii) occlusion penalties—augmented with small motion costs. Unlike metaheuristics that repeatedly sample without memory beyond a population state, QL is designed to enable (i) reuse of experience across similar scenes, (ii) online adaptation when targets or obstacles change, and (iii) anytime performance—delivering a usable orientation after every step. In this work we evaluate the first-order consequence of this design: the alignment between model physics and reward yields higher coverage than Harmony Search under matched evaluation budgets. The online-adaptation and cross-scene-transfer properties follow from the tabular value-function formulation but are not measured here; quantifying them is left to future work.

Contributions.

1. **$O^2ACD+QL$ framework:** We formulate sensor-orientation planning as an MDP whose reward is the marginal O^2ACD improvement, and instantiate a tabular Double Q-Learning agent with an ϵ -greedy/softmax exploration policy.
2. **Budget-fair evaluation vs. HS:** Under identical evaluation budgets (episodes $\times$ steps $\approx$ population $\times$ generations), QL outperforms HS across the tested scenarios and remains robust to changes in target count/size and FoV.

ORCID(s):

The target coverage problem is classified into l -coverage, k -coverage, and Q -coverage models [13]. In the first two models, the number of covering sensors for each target is fixed. More specifically, in the l -coverage model, each target should be covered by at least one sensor. However, in the k -coverage model, each target must be covered by at least $k \geq 2$ sensors. The problem focused on the Q -coverage model is different from those in the l - or k -coverage models. First, each target i of the Q -coverage model is monitored by $n_i \geq 1$ sensors. Second, the value of n_i varies for different targets, depending on their locations and coverage requirements.

While these existing models are sufficient for scalar sensors, they fail to model vector sensors accurately. This is because a vector sensor could only monitor a fraction of the target. Recently, Yang et al. [24] proposed a new coverage model for WVSNs, which is a direct application of the Q -coverage model and a quadruple viewing model proposed in [22]. Clearly, a directional sensor could only view a portion of the target that lies in its Field of View (FoV). This portion is called coverage degree (CD) and shown by solid red arcs in Fig. 1. In the CD -coverage model, the total coverage degree (TCD) of a target is defined simply as the sum of CDs for all covering sensors. It provides a rough estimation of the target's CD. A target is considered covered if its TCD is more than a predefined threshold. The threshold values are different for individual targets same as the original Q -coverage problem.

For typical WVSN applications such as surveillance cameras and environmental monitoring, the TCD model [24] is more accurate than l -coverage, k -coverage, and Q -coverage models. In addition, it allows us to convert the sophisticated coverage requirements of different targets into an optimization problem. The objective function of the resulting optimization problem is to maximize the number of covered targets by satisfying the TCD requirements of individual targets. Then, one could apply state-of-the-art optimization techniques to solve the coverage problem efficiently.

The TCD model has some important drawbacks that compromise its otherwise decent accuracy in modeling the relation between the directional sensors and under-surveillance target objects. First, it does not consider the possible intersection between visible parts of a target when the target is viewed by multiple sensors. As a result, the TCD measure overestimates the overall coverage of targets. This phenomenon, which is called CD overlap for the sake of brevity, is depicted in Fig. 1 (a). Here, sensors S_1 and S_2 cover target t partially. The CD overlap is accentuated in dense sensor networks where the sensor density is much more than the density of monitored targets. In such cases, each target will probably be monitored by multiple sensors. Therefore, the TCD model incorrectly overestimates the target's CD.

The TCD model also does not consider the interleaved occlusion between various targets. The impact of such a target occlusion problem is more severe when 1) the target density is higher than the sensor density, and 2) the target size is comparable to the sensors' FoVs. The first case is depicted in Fig. 1 (b) where the target density is higher than the sensor density. In terms of the S 's visibility, the target t_5 is completely blocked by other targets. Also, t_4 and t_2 are partially occluded by t_1 and t_3 , respectively. In Fig. 1 (c), the impact of target occlusion is ignored as in TCD. Here, the CD of covered targets is overestimated. The target size also impacts the TCD. The larger the target, the greater its occlusion impact on neighboring targets.

In this paper, we propose an enhanced CD model, called the overlap and occlusion aware coverage degree (O^2ACD) model, which considers the intersection between coverage degrees of adjacent sensors. It also finds obstructed parts of targets and so does not take these blocked parts into account when calculating CD. Furthermore, the TCD model uses a viewing state model [22] for modeling the relation between a sensor and a target. This model does not clearly define all possible cases. But, O^2ACD refines the viewing state model presented in [22] to further increase the modeling accuracy. Our simulation results indicate that O^2ACD is more accurate and efficient than the TCD model.

The rest of this paper is organized as follows. First, the related works are briefly reviewed in Section 2. Then, Section 3 presents the system model and problem definition. Section 4 describes O^2ACD in more detail. The simulation results are presented in Section 5. Finally, Section 6 concludes the paper.

2. Related Works

Yang et al. formulated the CD -coverage problem into a constraint optimization problem that can be solved by any optimization algorithm [24]. They have used the Harmony Search (HS) algorithm [9] to solve the problem. However, this approach suffers from three drawbacks. First, it does not consider the possible overlap between the coverage degrees of two adjacent sensors. Second, a target might be occluded partially or entirely by other targets. In that case, the occluded parts should be excluded from coverage degree calculations. Third, it does not consider physical obstacles such as walls, which reduce the observable part of a target.

Kyrkou et al. have developed a flexible uncertainty model to detect pedestrian behavior in cameras' FoV, which can be used in surveillance applications [11]. In their model, a camera determines the probability of detecting pedestrians by forming different FoVs related to a specific setting. This detection procedure is repeated for each pedestrian to find an appropriate configuration for individual cameras. After determining the cameras' configuration, they have used a camera network optimization algorithm to maximize overall detection efficiency. This algorithm uses the camera probability information in a flexible model to find the optimal network configuration.

In battery-free sensor wireless networks (BF-WSNs), energy is harvested from the environment instead of using batteries. Chen et al. have developed the first distributed algorithm to solve the minimum latency aggregation scheduling (MLAS) problem for BF-WSNs with coverage guarantee [5]. Alibeiki et al. have proposed a genetic algorithm (GA) to solve the k -coverage problem in directional sensor networks [1]. They used GA to solve the problem in an over-provisioned network and provide a balanced coverage in an under-provisioned network using a minimum number of directional sensors. A related genetic approach was later proposed for the target Q -coverage problem in over- and under-provisioned directional sensor networks [15].

Recently, the sequential vertex coloring-based sensor placement (SVC-SP) algorithm was proposed for the Q -coverage problem [3]. The SVC-SP algorithm determines the sensor requirements by dividing the specified target into independent subsets. A set of closer targets in the network forms an independent set, and a common sensor is set to monitor them. The random-based SVC-SP, cuckoo search-based SVC-SP, and GA-based SVC-SP algorithms have been used to find the optimal spot for sensors. Peng et al. have presented a new hybrid energy consumption method to increase the network lifetime called the new lifetime-enhancing method (NLEM) [17]. First, they solve the Q -coverage problem to find the optimum sensing direction for each sensor. Then, they utilize a new optimal cluster head selection algorithm and an inter-cluster communication algorithm to optimize the communication energy consumption.

3. System Model

A directional sensor has a finite angle of view, denoted by α . Suppose the sensing range of the sensor is r . Then, this sensor can cover a circular sector of area $\frac{1}{2}\alpha r^2$ (with α in radians) as depicted in Fig. 2 (a). Also, the sensor can adjust its FoV by changing its sensing direction. Each target is modeled by a circle with a radius r' . Here, r' is the radius of the smallest circle that encompasses the target shape. We assume that r' is smaller than r for all targets. We assume that all targets are stationary and have the same size. The locations of the sensors are fixed, too. For each target, only a proportion of its surface is observable by a sensor, as shown in Fig. 2 (a). The observed surface of the target is *arc-shaped*.

O²ACD is closely related to a previous model [22] when defining the relationship between a sensor and a target. In [22], the relative location of a target to a sensor is classified into four states: *white*, *black*, *grey*, and *half-grey*. A *black* target is completely unobservable by the sensor. In contrast, a *white* target completely lies in the sensor's FoV. For example, in Fig. 2b, t_2 and t_0 are black and white, respectively, with respect to S . For the sake of clarity, we have merged *grey* and *half-grey* targets into a new category called *partial-grey*, which also includes some new cases. For example, they did not consider a target observable if the center of the surrounding circle lies outside the sensor's FoV. Hence, in fig. 2c, t_4 , t_5 , and t_6 are considered black by [22] about S . Also, the target in Fig. 2a is wrongly considered as black by [22]. All of the targets in Fig. 2, except t_2 and t_0 are regarded as partial-grey in O²ACD.

3.1. Problem Statement

There are n targets $\mathbf{T} = \{t_1, t_2, \dots, t_n\}$ that should be covered by m sensors $\mathbf{S} = \{s_1, s_2, \dots, s_m\}$. The sensors are randomly deployed in the field with different working directions $\beta = \{\beta_1, \beta_2, \dots, \beta_m\}$. Here β_i is the angle between the direction of s_i (i in $[1, m]$) and x axis (see Fig. 2 (a)). For sensor s_i ($1 \leq i \leq m$) and target t_j ($1 \leq j \leq n$), suppose that the length of the observed arc on the target is l_{ij} . Thus, the coverage degree is defined as:

$$CD(s_i, t_j) = \frac{l_{ij}}{2\pi r'} \times 360, \quad (1)$$

where CD is measured in degrees. Accordingly, the overall coverage degree of t_j i.e. $TCD(t_j)$ is defined as:

$$TCD(t_j) = \sum_{i=1}^m CD(s_i, t_j) = \frac{360}{2\pi r'} \sum_{i=1}^m l_{ij}. \quad (2)$$

A target t_j is said to be detected if $TCD(t_j)$ is more than the predefined threshold τ_j . Now, the CD -coverage problem could be stated as an optimization problem as follows.

Problem 1. For a sensor set S , a target set T , and a coverage requirements vector $\mathbf{C} = \{\tau_1, \tau_2, \dots, \tau_n\}$, find the optimum working direction β to maximize the number of detected targets as the objective function.

The above formulation has two nice properties. First, it converts a non-scalar coverage problem into a scalar equivalent problem (i.e., Q -coverage). Second, the scalar coverage parameter (i.e. τ) could accommodate other desirable requirements such as coverage priorities, energy-saving, data aggregation, etc., as well.

3.2. Q-Learning for Orientation Planning

Besides the population-based metaheuristics, we plan the sensor orientations with a tabular reinforcement-learning agent based on Q-Learning [23]. The orientation-planning task is cast as a multi-agent Markov decision process (MDP) in which each of the m directional sensors is an independent learning agent that adjusts its working direction β ; all agents share the same reward signal derived from the O^2ACD coverage model, so learning is cooperative. The formulation below follows the implementation exactly.

State. The state of agent i is the quantization of its current orientation $\beta_i \in [0, 2\pi)$ into $N_s = 72$ equal bins of 5° ,

$$s_i = \left\lfloor \frac{\beta_i}{2\pi} N_s \right\rfloor + 1, \quad s_i \in \{1, \dots, N_s\}.$$

Action. Each agent chooses one of $N_a = 5$ discrete rotations

$$a_i \in \mathcal{A} = \{-30^\circ, -15^\circ, 0^\circ, +15^\circ, +30^\circ\},$$

and its orientation is updated by $\beta_i \leftarrow (\beta_i + a_i) \bmod 2\pi$.

Reward. After a joint action, the environment is re-evaluated with the O^2ACD model. Let Δu_i be the change in the unique (non-overlapping) coverage contributed by sensor i , u_i its current total unique coverage, o_i its current overlap with other sensors, ΔU the change in the total coverage, and ΔT the change in the number of covered targets. The per-agent reward is

$$r_i = \begin{cases} -\pi, & \text{if } u_i \leq 0 \text{ (sensor covers nothing),} \\ w_1 \Delta u_i + w_2 o_i + w_3 \Delta U + w_4 \Delta T, & \text{otherwise,} \end{cases}$$

with weights $(w_1, w_2, w_3, w_4) = (1.0, -0.4, 0.15, 1.5)$. The positive terms reward added unique coverage and newly covered targets, while the negative weight w_2 penalizes redundant overlap between sensors—precisely the redundancy the O^2ACD model is designed to eliminate.

Update rule. To reduce the maximization bias of standard Q-Learning, each agent keeps a pair of action-value tables and is trained with Double Q-Learning [10]. At every step, with probability $1/2$ one table is updated using the other for bootstrapping:

$$Q_1(s, a) \leftarrow Q_1(s, a) + \alpha [r + \gamma Q_2(s', \arg \max_{a'} Q_1(s', a')) - Q_1(s, a)],$$

and symmetrically for Q_2 otherwise. Actions are selected from the averaged table $\bar{Q} = \frac{1}{2}(Q_1 + Q_2)$ with an ϵ -greedy policy whose exploitation branch draws from a softmax (Boltzmann) distribution with temperature θ (a distinct symbol from the coverage threshold τ used in Sec. 4).

Hyperparameters. The discount factor is $\gamma = 0.97$. The learning rate is annealed per episode as $\alpha_e = \alpha_0 / \sqrt{e}$ with $\alpha_0 = 0.02$. The exploration rate starts at $\epsilon = 0.25$ and decays by a factor of 0.995 each episode down to a floor of 0.02; the softmax temperature starts at $\theta = 0.25$ and decays by 0.998 per episode. The agent runs for $E = n\text{Gen}$ episodes of $K = n\text{Pop}$ steps each, so that the total number of coverage evaluations matches the population $\times$ generations budget of the metaheuristics, ensuring a budget-fair comparison.

Initialization and termination. The tables Q_1, Q_2 are initialized with negligible random noise to break ties. At the start of each episode the sensor orientations are re-initialized uniformly at random over $[0, 2\pi)$. At the end of every episode a greedy pass (selecting $\arg \max_a \bar{Q}$ for each sensor) is evaluated, and the orientation vector achieving the largest number of covered targets across all episodes is returned. Training terminates when the fixed episode budget E is exhausted. The procedure is summarized in Algorithm 1 (Q-LEARNING).

Algorithm 1 Q-LEARNING for multi-agent orientation planning**Input:** sensors $\mathcal{S}$ (m agents), targets $\mathcal{T}$, episodes E , steps K **Output:** best orientation vector β^*

```

1: procedure Q-LEARNING( $\mathcal{S}, \mathcal{T}, E, K$ )
2:   initialize  $Q_1(s, a, i), Q_2(s, a, i) \leftarrow$  small random noise ▷ one table pair per agent
3:    $C^* \leftarrow -\infty$ ;  $\epsilon \leftarrow 0.25$ ;  $\theta \leftarrow 0.25$ 
4:   for  $e \leftarrow 1$  to  $E$  do
5:     randomize orientations  $\beta$ ;  $s \leftarrow \text{ENCODE}(\beta)$ 
6:      $\alpha \leftarrow 0.02/\sqrt{e}$  ▷ anneal learning rate
7:     evaluate  $O^2\text{ACD}$  at  $\beta$ ; cache coverage
8:     for  $k \leftarrow 1$  to  $K$  do
9:       for each agent  $i$  do
10:         $a_i \leftarrow \text{SELECT}(\bar{Q}(\cdot, \cdot, i), s_i, \theta, \epsilon)$  ▷  $\epsilon$ -greedy over softmax
11:      end for
12:       $\beta \leftarrow (\beta + \mathbf{a}) \bmod 2\pi$ 
13:      re-evaluate  $O^2\text{ACD}$ ; compute rewards  $\mathbf{r}$  ▷  $-\pi$  if a sensor covers nothing
14:       $s' \leftarrow \text{ENCODE}(\beta)$ 
15:      for each agent  $i$  do
16:        if  $\text{RAND}() < 0.5$  then
17:           $a^\dagger \leftarrow \arg \max_{a'} Q_1(s'_i, a', i)$ 
18:           $Q_1(s_i, a_i, i) += \alpha [r_i + \gamma Q_2(s'_i, a^\dagger, i) - Q_1(s_i, a_i, i)]$ 
19:        else
20:           $a^\dagger \leftarrow \arg \max_{a'} Q_2(s'_i, a', i)$ 
21:           $Q_2(s_i, a_i, i) += \alpha [r_i + \gamma Q_1(s'_i, a^\dagger, i) - Q_2(s_i, a_i, i)]$ 
22:        end if
23:      end for
24:       $s \leftarrow s'$ 
25:    end for
26:     $\epsilon \leftarrow \max(0.02, 0.995 \epsilon)$ ;  $\theta \leftarrow \max(10^{-3}, 0.998 \theta)$  ▷ decay schedules
27:     $\beta_g \leftarrow \text{GREEDYPASS}(\beta, \bar{Q})$ ;  $C \leftarrow$  covered targets at  $\beta_g$ 
28:    if  $C > C^*$  then
29:       $C^* \leftarrow C$ ;  $\beta^* \leftarrow \beta_g$  ▷ keep best-so-far
30:    end if
31:  end for
32:  return  $\beta^*$ 
33: end procedure

```

4. $O^2\text{ACD}$

Our proposed model considers coverage overlap, target occlusion, and physical obstacles at once. We refine the problem definition to apply our optimized model as follows. First we model l_{ij} in Eq. (1) as an interval such as $[a_{ij}, b_{ij}]$ so that $0 \leq a_{ij} < b_{ij} \leq 2\pi$. Clearly, for a set of disjoint intervals, the length of their union is the sum of their lengths. Therefore, Eq. (2) only holds for a set of disjoint intervals. We modify Eq. (2) as follows to accommodate overlapping intervals:

$$TCD_u(t_j) = \frac{360}{2\pi r'} \left| \bigcup_{i=1}^m [a_{ij}, b_{ij}] \right|, \quad (3)$$

Here, the union operator $\cup$ takes the overlapping intervals into TCD's calculation only once. It is worth noting that finding the union of n overlapping intervals is an $O(n \log(n))$ problem [8].

Since our model is sensitive to occlusion between targets, we can easily modify the model to consider the existence of physical obstacles in the field as follows. First, it is not required to cover an obstacle with the sensors. Therefore,

Table 1

Coverage ratio (mean over 50 trials) of the O²ACD (Union) model under the QL and HS optimizers as the number of targets n varies, at the baseline setting ($\alpha=45^\circ$, $\tau=240^\circ$). Bold = best in row.

Scenario (n , α , τ)	QL (mean)	HS (mean)	Δ (abs, QL–HS)
(50, 45° , 240°)	0.644	0.617	0.027
(70, 45° , 240°)	0.569	0.537	0.032
(120, 45° , 240°)	0.466	0.452	0.014

we model an obstacle as a target whose total coverage degree is zero. The obstacle is a target with $\tau = 0^\circ$, and radius r'' . Here, r'' is the radius of the smallest circle that encircles the obstacle (see Fig. 3). Now, O²ACD treats all physical obstacles as targets and considers the resulting occlusion. Finally, we formulate the CD-coverage problem into a constraint optimization problem in the presence of physical obstacles as follows:

Problem 2. For a sensor set $\mathbf{S} = \{s_1, s_2, \dots, s_m\}$, a target set $\mathbf{T} = \{t_1, t_2, \dots, t_n\}$ with a coverage requirements vector $\mathbf{C} = \{\tau_1, \tau_2, \dots, \tau_n\}$, and an obstacle set $\mathbf{O} = \{o_1, o_2, \dots, o_k\}$, find the optimum working direction β to maximize the number of detected targets:

$$\text{Maximize } \sum_{i=1}^n c_i, \quad c_i = \begin{cases} 1, & TCD_u(t_i) \geq \tau_i \\ 0, & TCD_u(t_i) < \tau_i \end{cases} \quad (4)$$

where c_i indicates whether target t_i is covered by a given set of sensors (i.e., its union coverage degree $TCD_u(t_i)$ meets its requirement τ_i).

5. Simulation Results

Under budget parity with HS, Q-Learning achieves higher coverage across all tested scenarios. The absolute margin over HS in the baseline setting is modest (0.014–0.032 in coverage ratio; Table 1) and does not vary monotonically with the number of targets.

Table 1 reports the coverage ratio (mean over 50 trials) of the O²ACD (Union) model for the QL and HS optimizers, as the number of targets n varies at the baseline setting ($\alpha=45^\circ$, $\tau=240^\circ$). Bold = best.

We have used MATLAB to evaluate the O²ACD model against the CD-coverage model. The size of the simulation area is 750m×750m. The default values for sensing range r and viewing angle α are 120m and 45° . Besides, m (i.e., the number of sensors) and n (the number of targets) are 75 and 50, respectively. We repeated each simulation scenario 50 times with random placement of sensors and targets. Furthermore, initial settings, such as the placement of sensors and targets, are the same for evaluated models. The coverage ratio is defined as the ratio between covered targets and the number of targets n .

In all plots, TCD represents the CD-coverage model [24]. O¹ACD is O²ACD without target occlusion. O²ACD is the proposed model, which considers both interval intersection and target occlusion. For a fair comparison, TCD is equipped with our enhanced target viewing model, which considers *black*, *grey*, and *partial-grey* targets. We have assumed that τ is equal for all targets. The value of τ is fixed in all simulation experiments (i.e., 240°) except the last one. Also, we have implemented HS[9], PSO[12], DE[18], ACO[7], and CA[20] optimization methods for evaluating O²ACD, O¹ACD, and TCD models.

5.1. Evaluated Models

In Fig. 4, the coverage ratio is depicted against the number of targets from $n=40$ to $n=120$. The coverage ratio decreases with n for a fixed number of sensors. Fig. 4 reports one panel per optimization method (QL, HS, PSO, DE, ACO, and CA), each showing the three coverage models. Among the evaluated approaches, QL achieves the highest coverage ratio in every panel, with HS the strongest of the metaheuristic baselines; the remaining methods trail behind, and PSO in particular degrades sharply as n grows in this sweep (its panel falls to well below 10% coverage at large n). In these panels O²ACD reports a higher coverage ratio than TCD, and slightly higher than O¹ACD. In other words, the impact of interval overlap is higher than target occlusion. Furthermore, for a fixed number of sensors, the number

of visible targets is limited. Therefore, when n (number of targets) increases, the coverage ratio decreases. Also, the number of occluded targets increases with n . This is also another reason behind the coverage ratio reduction.

Fig. 5 plots the impact of target size (i.e. r') on the coverage ratio. The visible degree of a target, i.e. $\frac{l_{ij}}{2\pi r'} \times 360$ decreases when r' increases for a fixed sensing range r . Therefore, the coverage ratio decreases with r' . Again, O^2ACD reports a higher coverage ratio than TCD across this sweep. More interestingly, the gap between O^2ACD and O^1ACD is increasing with r' since the occlusion possibility is higher for larger targets. As in Fig. 4, each panel corresponds to one optimization method; QL again attains the highest coverage ratio, followed by HS, while ACO and CA yield the lowest coverage among the compared methods.

Fig. 6 depicts the coverage ratio vs. viewing angle α . All methods linearly improve with α , which is completely expected. Improving the viewing angle directly impacts the sensors' FoV. As a result, the sensors can monitor a higher number of targets. Again, the coverage ratio of O^2ACD is higher than TCD across this sweep.

Finally, Fig. 7 depicts the coverage ratio against the threshold τ . Here, the coverage ratio decreases sharply when τ increases: the larger τ is, the smaller the number of coverable targets. In this sweep the ordering of the models differs from the previous figures: O^2ACD reports a *lower* coverage ratio than TCD, with O^1ACD in between. Across the sweeps, O^2ACD and O^1ACD differ only slightly, indicating that interval overlap contributes more to the correction than target occlusion at these settings; the occlusion term becomes more pronounced for larger targets and dense networks where n is large relative to m .

5.2. Evaluated AI Methods

In this section, we compare the performance of the evaluated optimization methods using our models. To make the comparison across scenarios explicit, the results in Fig. 8 are normalized to QL, so that QL corresponds to unity and every other method is reported as a fraction of the QL coverage. Evidently, QL attains the best coverage in all scenarios. HS is the closest competitor, typically reaching about 86–99% of the QL coverage. DE is the next best method, at roughly 58–71% of QL. CA and ACO form a lower band, generally below 55% of QL, and their relative order varies with the scenario, while PSO is the weakest method overall (falling to roughly 12–15% of QL in the target-count sweep). This confirms that the learning-based planner exploits the O^2ACD model more effectively than the metaheuristic baselines.

6. Conclusion

We introduced a new coverage-degree model for WVSNs that measures, for each target, the union of its visible arc intervals from all covering sensors and explicitly subtracts occluded portions. By eliminating double-counting and removing line-of-sight occlusions from the score, the model is more faithful than CD/TCD and, when optimized, increases the number of truly covered targets.

Building on this foundation, we coupled the model (O^2ACD) with tabular Q-Learning. Under matched evaluation budgets, Q-Learning consistently outperforms Harmony Search (HS) while preserving the model's physical correctness. Because the policy is a tabular value function over orientation actions, it can in principle adapt online and warm-start from related scenes; a quantitative evaluation of these properties is left to future work. The framework naturally extends to function approximation (e.g., deep Q-networks [14] and actor–critic methods) and dynamic environments.

Why Q-Learning helps. Q-Learning exploits temporal credit assignment: it learns which rotation directions yield marginal gains in the union-of-arc coverage given local geometry, thereby avoiding redundant overlaps that HS repeatedly rediscovers. A synchronous variant further reduces reward noise by evaluating all sensors together each step, aligning the reward with their joint effect on O^2ACD .

Practicality. With discretized actions, tabular Q-Learning keeps computation on par with HS and offers anytime control (a usable orientation after every step); warm-starting from prior scenes is a property of the tabular formulation that we have not yet evaluated empirically.

Limitations and next steps. Discrete actions and tabular states may not scale to high-DoF PTZ cameras or complex occluders. Replacing the table with function approximators and moving to finer or continuous control are natural extensions, provided O^2ACD remains the reward signal.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a generative artificial intelligence large language model tool in order to improve the readability and language of the manuscript and to assist with editing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

References

- [1] Alibeiki, A., Motameni, H., Mohamadi, H., 2021. A new genetic-based approach for maximizing network lifetime in directional sensor networks with adjustable sensing ranges. *Future Generation Computer Systems* 119, 102–112.
- [2] Almalkawi, I.T., Al-Karaki, J.N., Alsmadi, I., 2024. Intelligent camera network management for visual sensor networks. *Journal of Network and Computer Applications*.
- [3] Arivudainambi, D., Pavithra, R., 2021. Vertex coloring approach for Q-coverage problem in wireless sensor network. *Journal of Intelligent & Fuzzy Systems* 41, 5541–5551. doi:10.3233/JIFS-191795.
- [4] Cai, Y., Li, X., Zhang, Y., 2023. Coverage optimization of directional sensor networks based on deep reinforcement learning. *Sensors* 23, 3009.
- [5] Chen, Q., Cai, Z., Cheng, L., Gao, H., 2019. Low-latency data aggregation scheduling for battery-free wireless sensor networks. *IEEE Transactions on Mobile Computing*.
- [6] Din, A., Ismail, M.Y., Shah, B., Babar, M., Ali, F., Baig, S.A., 2022. A deep reinforcement learning-based multi-agent area coverage control for smart agriculture. *Computers and Electrical Engineering* 101, 108089. doi:10.1016/j.compeleceng.2022.108089.
- [7] Dorigo, M., Birattari, M., Stützle, T., 2006. Ant colony optimization. *IEEE Computational Intelligence Magazine* 1, 28–39. doi:10.1109/MCI.2006.329691.
- [8] Fredman, M.L., Weide, B., 1978. On the complexity of computing the measure of $\bigcup [a_i, b_i]$. *Communications of the ACM* 21, 540–544. doi:10.1145/359545.359553.
- [9] Geem, Z.W., Kim, J.H., Loganathan, G.V., 2001. A new heuristic optimization algorithm: Harmony search. *Simulation* 76, 60–68. doi:10.1177/003754970107600201.
- [10] van Hasselt, H., Guez, A., Silver, D., 2016. Deep reinforcement learning with double Q-learning, in: *Proceedings of the AAAI Conference on Artificial Intelligence*, pp. 2094–2100. doi:10.1609/aaai.v30i1.10295.
- [11] Kyrkou, C., Laoudias, C., Theocharides, T., Panayiotou, C.G., Polycarpou, M.M., 2018. Optimizing the detection performance of smart camera networks through a probabilistic image-based model. *IEEE Transactions on Circuits and Systems for Video Technology* 28, 1197–1211.
- [12] Marini, F., Walczak, B., 2015. Particle swarm optimization (PSO), a tutorial. *Chemometrics and Intelligent Laboratory Systems* 149, 153–165. doi:10.1016/j.chemolab.2015.08.020.
- [13] Mavrinac, A., Chen, X., 2013. Modeling coverage in camera networks: A survey. *International Journal of Computer Vision* 101, 205–226. doi:10.1007/s11263-012-0587-7.
- [14] Mnih, V., Kavukcuoglu, K., Silver, D., Rusu, A.A., Veness, J., Bellemare, M.G., Graves, A., Riedmiller, M., Fidjeland, A.K., Ostrovski, G., et al., 2015. Human-level control through deep reinforcement learning. *Nature* 518, 529–533. doi:10.1038/nature14236.
- [15] Mottaki, F., Motameni, H., Mohamadi, H., 2022. A genetic algorithm-based approach for solving the target Q-coverage problem in over and under provisioned directional sensor networks. *Wireless Networks*.
- [16] Mou, Z., Gao, Y., Liu, J., Gao, F., Chen, Z., Ho, P.H., 2021. Deep reinforcement learning based three-dimensional area coverage with UAV swarm. *IEEE Journal on Selected Areas in Communications* 39, 3160–3176. doi:10.1109/JSAC.2021.3088718.
- [17] Peng, J., Chen, B., Liu, A., 2020. A new lifetime-enhancing method for Q-coverage in wireless sensor networks. *IEEE Access*.
- [18] Price, K.V., 2013. Differential evolution, in: *Handbook of Optimization*. Springer. volume 38 of *Intelligent Systems Reference Library*, pp. 187–214. doi:10.1007/978-3-642-30504-7_8.
- [19] Pushpalatha, S., Shivaprakasha, K.S., 2025. Optimizing coverage in wireless sensor networks using reinforcement learning. *Wireless Personal Communications*.
- [20] Reynolds, R.G., 1994. An introduction to cultural algorithms, in: *Proceedings of the Third Annual Conference on Evolutionary Programming*, World Scientific. pp. 131–139.
- [21] Soltani, N., Rahmani, A.M., 2025. Adaptive multi-agent reinforcement learning for vision-based coverage control. *Engineering Applications of Artificial Intelligence*.
- [22] Wang, Y., Cao, G., 2011. On full-view coverage in camera sensor networks. *Proceedings IEEE INFOCOM*, 1781–1789. doi:10.1109/INFOCOM.2011.5934977.
- [23] Watkins, C.J.C.H., Dayan, P., 1992. Q-learning. *Machine Learning* 8, 279–292. doi:10.1007/BF00992698.
- [24] Yang, H., Li, X., Wang, Z., Yu, W., Huang, B., 2019. A coverage degree-coverage model in wireless visual sensor networks. *IEEE Access* 7, 13264–13274. doi:10.1109/ACCESS.2019.2892190.
- [25] Zhou, Y., Li, W., Zhang, H., 2025. Adaptive deployment of wireless sensor networks based on deep reinforcement learning. *Ad Hoc Networks* 166, 103672.

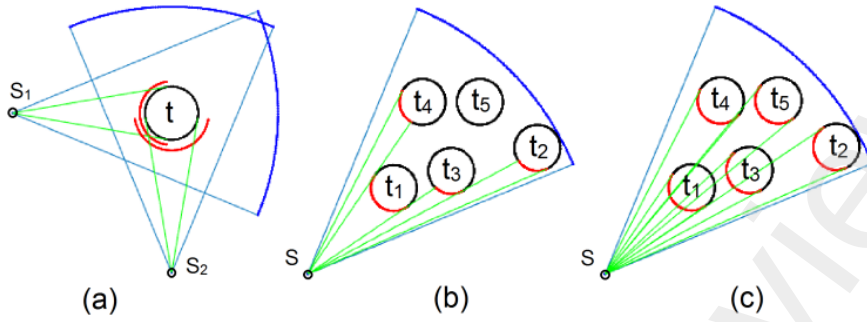


Figure 1: The coverage overlap (a) and target occlusion (b) problems. (c) depicts the coverage when ignoring the target occlusion problem.

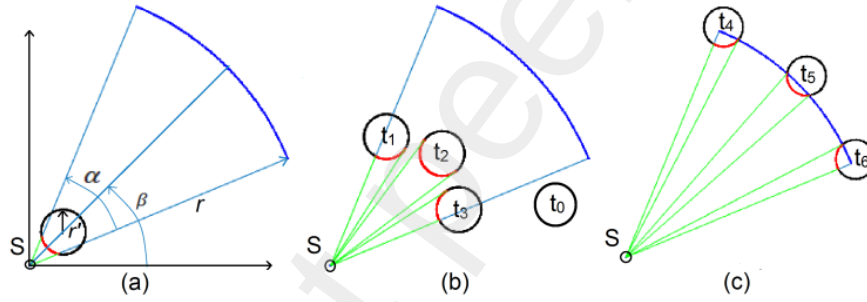


Figure 2: The relation between a sensor and covered targets.



Figure 3: O^2ACD treats an obstacle as a target with zero coverage-degree requirements.

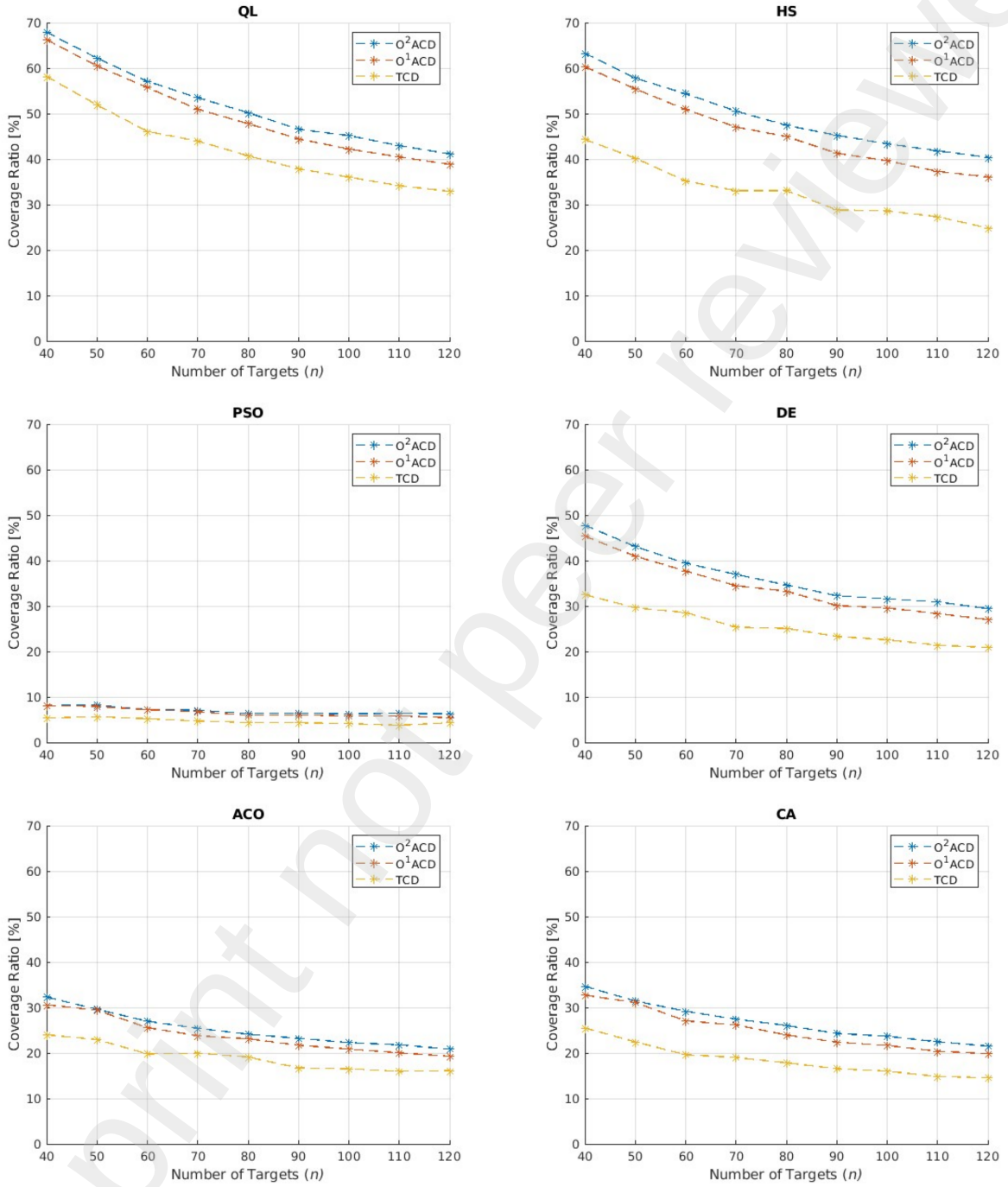


Figure 4: Coverage ratio comparison of the O^2ACD , O^1ACD , and TCD models vs. the number of targets (n) for a fixed number of sensors, one panel per optimization method.

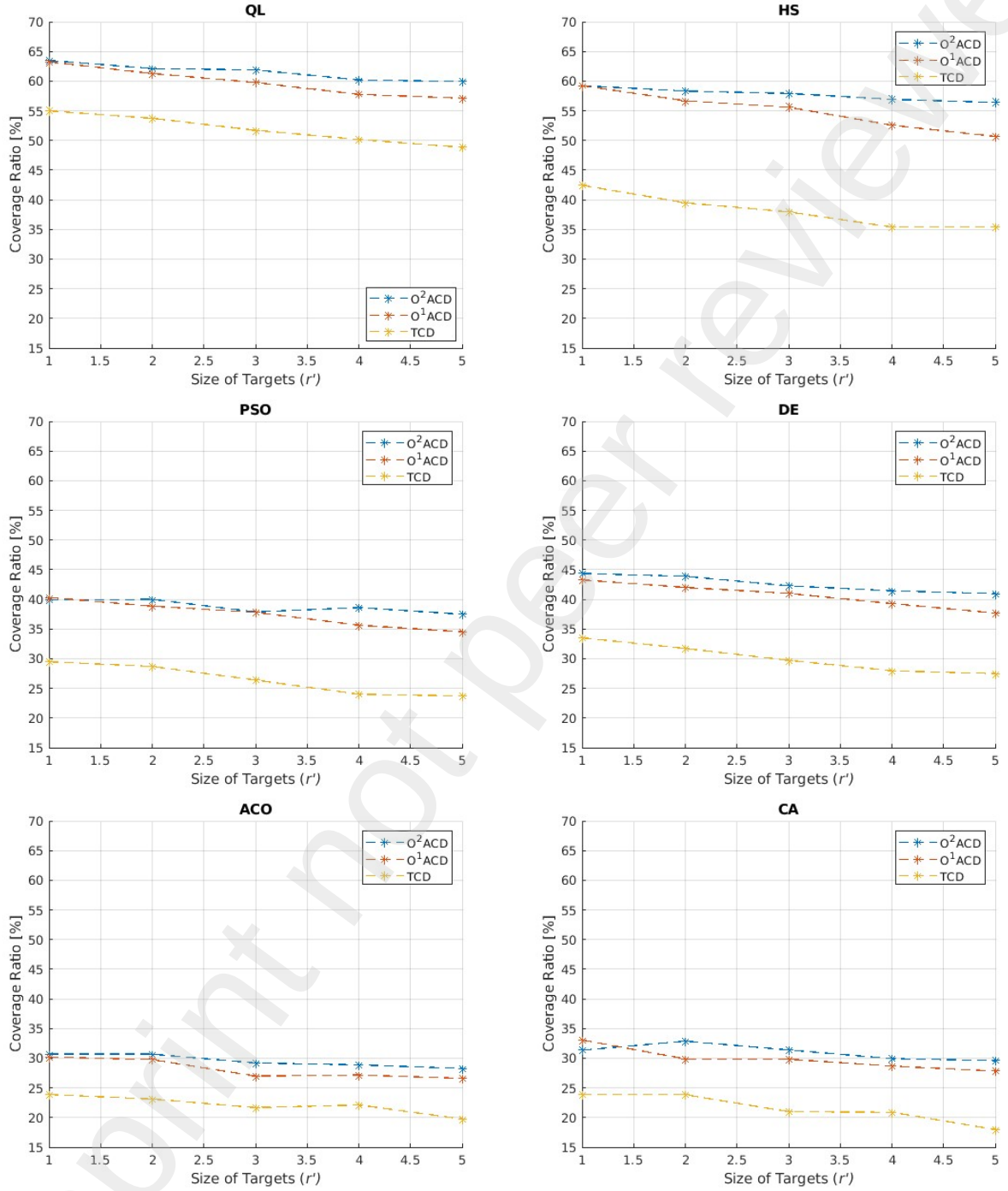


Figure 5: Coverage ratio vs. target size r' , one panel per optimization method. The target size has a reverse effect on target observability. The number of sensors (m) and targets (n) is fixed.

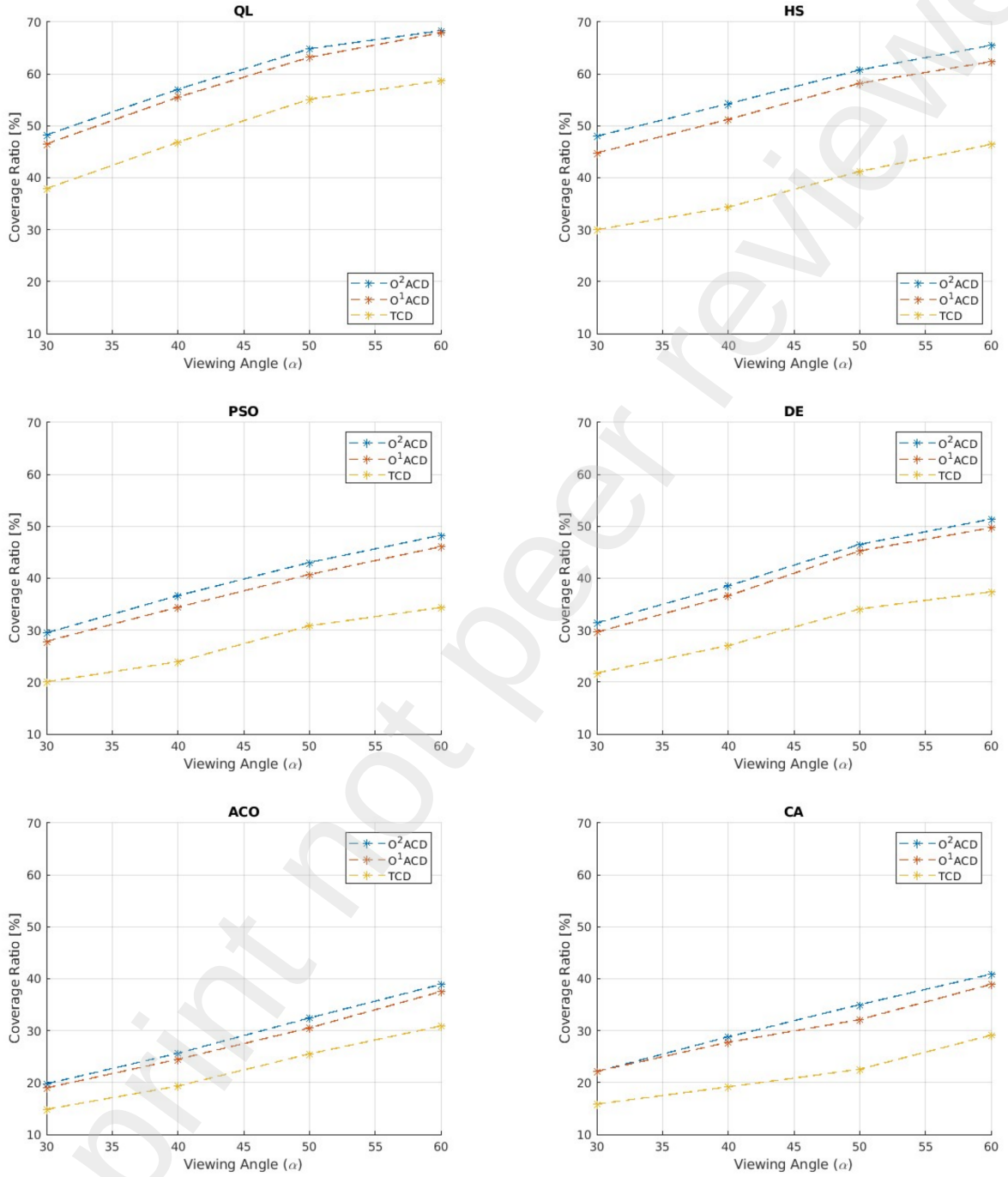


Figure 6: Coverage ratio vs. viewing angle α , one panel per optimization method. The coverage ratio increases with viewing angle α for a fixed m and n .

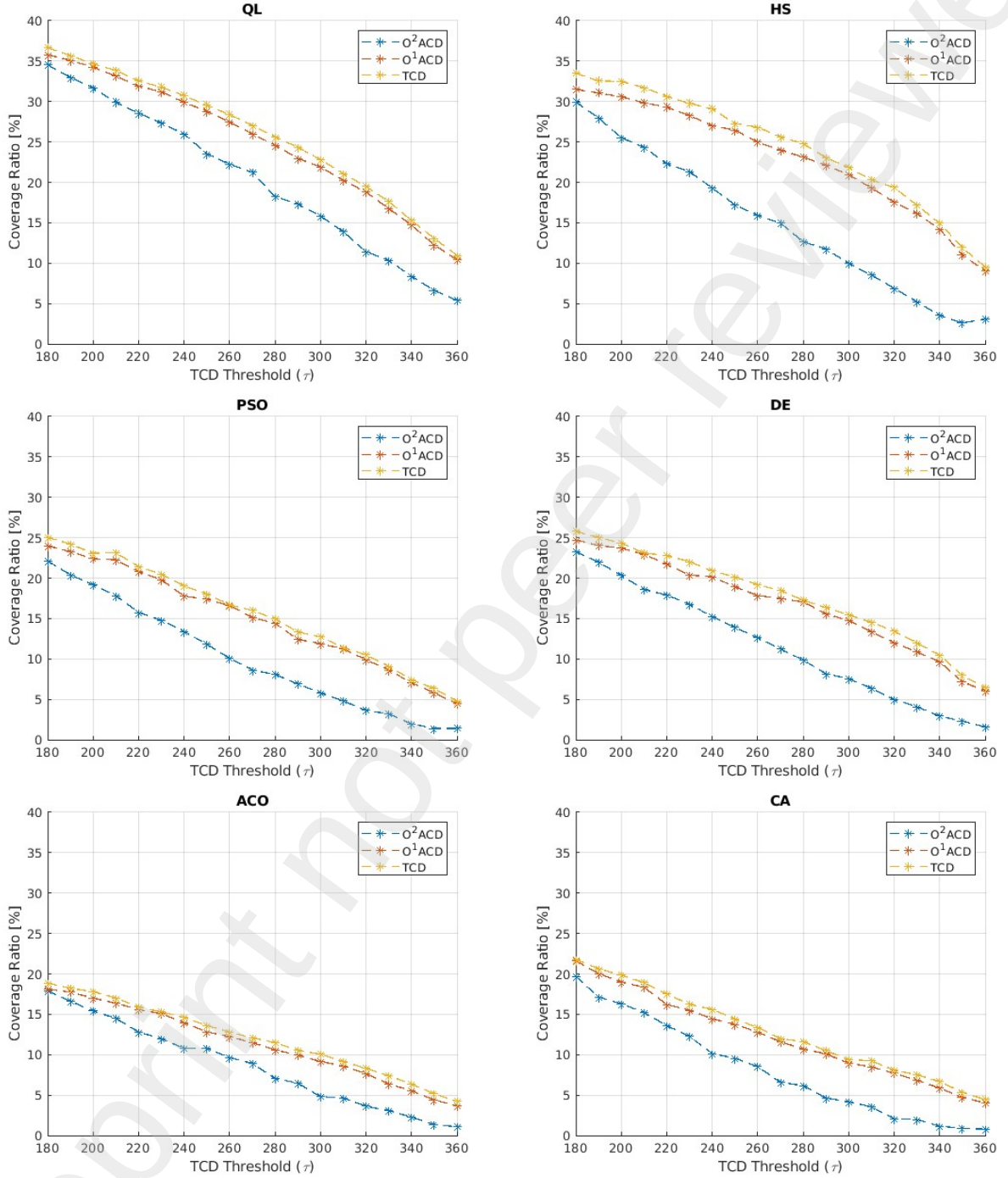


Figure 7: Coverage ratio vs. TCD threshold τ , one panel per optimization method. The coverage ratio decreases with τ for a fixed m and n .

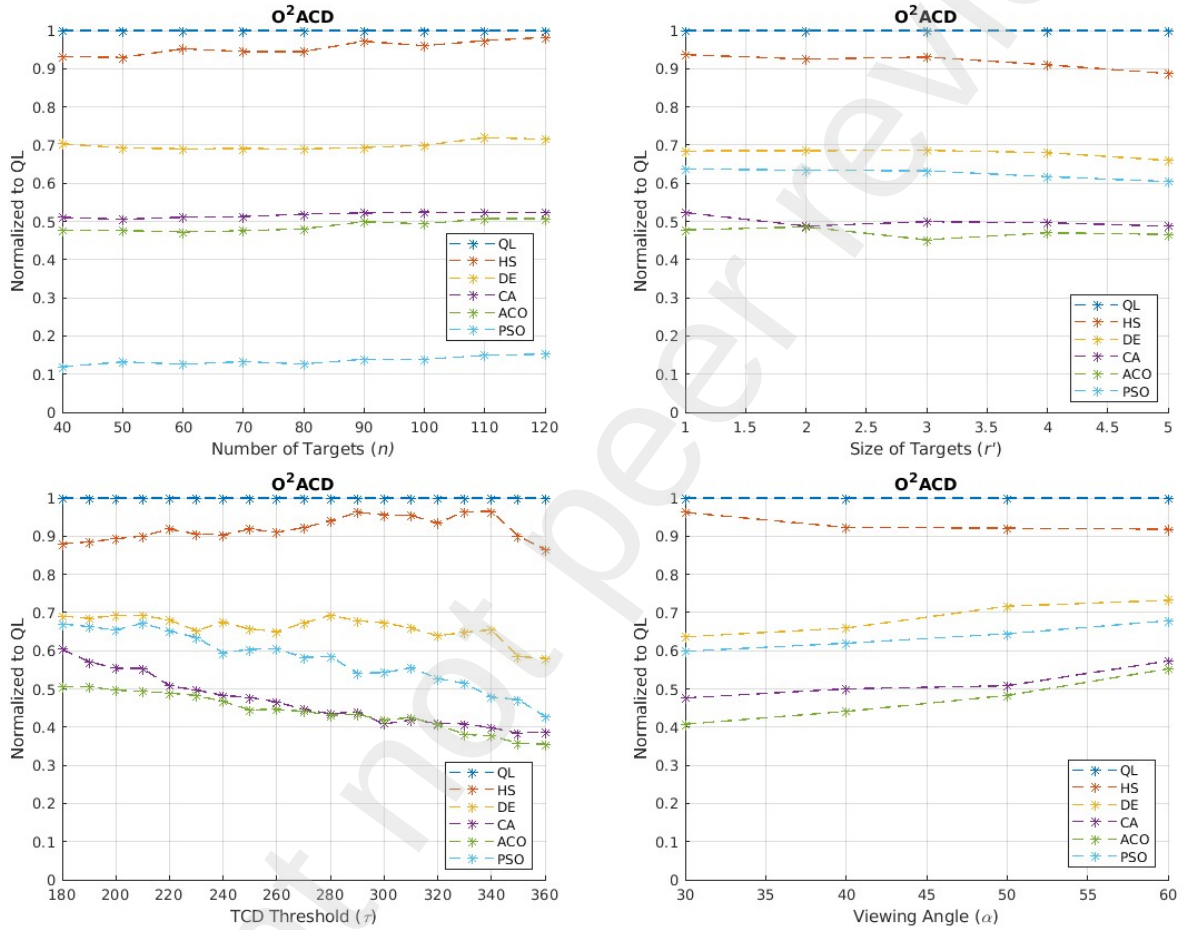


Figure 8: Performance of the optimization methods normalized to QL. In all cases, QL attains the best coverage; HS is the closest competitor (about 86–99% of QL), while the remaining methods fall well below QL.